Which Post-Quantum Signature Modes Fit Broadcast Budgets? A Cross-Domain Scoring of Authentication Modes for Handshake-Free Cyber-Physical Broadcasts

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DRAFT – PROVISIONAL SMOKE-TEST NUMBERS

All measurements come from smoke runs on a shared cloud VM (machine tag `cloud-smoke`, 2026-11-11, commit `3cf20df`) with 1000 iterations (20 for slow schemes). They are not publication numbers. Every value is defined in `numbers.tex` and traced to `results/summary.md`; open items are marked TODO-RESULT or [TODO].

***Abstract*—Broadcast channels in cyber-physical systems (vehicle-to-vehicle messages, drone Remote ID, satellite navigation message authentication, cellular system information, DNS responses) cannot run a handshake, and each has a hard per-message byte budget. Prior post-quantum (PQ) studies fit signatures to one such budget at a time. We score one set of authentication modes (full signature, signature with certificate, fragmented signature, a PQ-anchored symmetric key chain, and a classical/PQ hybrid) against five budget profiles from five domains, using signature sizes and timings measured with liboqs 0.16.0 on x86. In this provisional data, a full PQ signature fits in 104 of 1105 (parameter set, profile) pairs, all of them UOV, SNOVA, MAYO or Falcon; no parameter set fits when a same-scheme certificate is attached. A loss model comparing the PQ-anchored chain to fragmented signatures at 0–30% i.i.d. loss favours the chain on bytes per verified message and 95th-percentile time to authenticate, but not on verified share at low loss, and depends on assumed chain parameters; we report it as inconclusive. On x86, signature size, not the sign/verify timing tail, is the binding constraint for most families; constrained microcontrollers were not measured.**

***Index Terms*—post-quantum signatures, broadcast authentication, cyber-physical systems, V2X, Remote ID, GNSS, 5G**

I. Introduction

Many safety-relevant cyber-physical systems authenticate by broadcast. A vehicle sends a basic safety message every 100 ms [1], [2]; a drone broadcasts Remote ID over Bluetooth frames with 25-octet payloads [3]; Galileo OSNMA spreads authentication data over navigation message blocks [4]; a 5G base station broadcasts SIB1 every 160 ms [5]. In each case the receiver is passive, there is no handshake, and the channel imposes a fixed byte budget per message. Today these channels rely on classical signatures or on symmetric schemes. Replacing them with standardized post-quantum signatures [TODO: cite: FIPS 204 / FIPS 205 / FN-DSA draft] is hard because PQ signatures and public keys are often an order of magnitude larger.

Prior work answers the fit question one domain at a time: for C-V2X [6]–[8], OSNMA [4], 5G SIB [5] and DNSSEC [9]. Multi-algorithm comparisons inside one domain also exist, e.g. for V2X [10] and as a review of ITS [2]. What is missing is a common method that scores the same *authentication modes* against several handshake-free budgets, so that a designer can see whether a mode that works in one domain transfers to another. Our contribution is this cross-domain, mode-level scoring. It is not breadth of algorithms: the number of parameter sets we benchmark is a by-product of using liboqs.

Contributions.

- A byte-accounting method for five authentication modes (full, signature+certificate, fragmented, PQ-anchored chain, hybrid) and five broadcast budget profiles taken from published specifications and papers (C-V2X, Remote ID, OSNMA, 5G SIB1, DNS) (Sec. III).
- A scoring of 221 PQ parameter sets measured with liboqs against these profiles (Q1). Only UOV, SNOVA, MAYO and two Falcon sets fit any profile with a full signature; no set fits with a certificate.
- A seeded loss model comparing the PQ-anchored chain with fragmented signatures at 0–30% loss (Q2). The chain wins on bytes and latency in all cells but not on verified share at low loss; the outcome is assumption-sensitive and we call it inconclusive.
- A size-versus-timing-tail comparison against 3 ms and 100 ms deadlines (Q3), on x86 only.

All results in this draft are provisional smoke-test runs; limitations are collected in Sec. V.

II. Background and Related Work

Single-domain size fit. Twardokus et al. [6] argue that the 437 B C-V2X payload limit makes direct PQ signatures impractical and propose a partially hybrid V2V protocol. Abrar et al. [7] and Chen and Lin [8] study PQ signatures and hybrid certificates in C-V2X and its security credential management system [11]. Junquera-Sánchez et al. [4] assess PQ options for Galileo OSNMA. Goertzen and Stebila [9] carry PQ signatures in DNSSEC over the 1232 B UDP limit by request-based fragmentation; Rawat and Jhanwar [12] propose a QNAME-based variant (we have read only its abstract). These works each fix one budget.

Multi-algorithm, single domain. Yoshizawa et al. [10] compare NIST and additional-round signatures for V2X messages and compress them with zero-knowledge proofs. Mamun et al. [2] review PQ cryptography for ITS. Nabasirye et al. [13] describe a PQC benchmarking framework on dual-core Xtensa hardware; we have read only the abstract and do not rely on its numbers.

PQ-anchored and symmetric broadcast authentication. TESLA-style delayed key disclosure [TODO: cite: TESLA, Perrig et al.] authenticates broadcasts with short MACs and one-way key chains, at the cost of loose time synchronization. TBRD [14] applies it to Remote ID. Darzi et al. [5] anchor a TESLA-style HMAC chain with one MAYO signature per epoch so that SIB1 authentication fits in one 372 B message with 324 B of cryptographic overhead, where direct ML-DSA or FN-DSA signatures need 34 or 13 SIB1 fragments. Hussain et al. [15] use a fast MAC for IEC 61869-9 sampled values under a 3 ms timing limit. Stateful and few-time hash-based schemes [16], [17] give fast signing but have finite signing capacity; we do not benchmark them.

Hybrids. RFC 9955 [18] defines vocabulary for hybrid signature properties. Surveys of hybrid deployment in TLS/QUIC [19] and compact non-separable hybrid constructions [20] exist; we have read only their abstracts. We model the simplest hybrid, concatenation of a classical and a PQ signature.

Microcontroller benchmarks. Chhetri et al. [21] report ML-DSA latency, rejection-sampling variance and memory on an RP2040 (Cortex-M0+). Further Cortex-M work covers MAYO [22], [23], SQIsign verification [24] and Falcon [25], [26], and pqm4 benchmarks additional signatures [27]; we have read only abstracts of these and cite them as pointers, not for numbers. Strenzke [28] analyses, per its abstract, how signature memory and bandwidth scale with message size. None of these maps results onto broadcast budgets.

Feasibility envelopes. The closest framing we know is Hyoung et al. [29], who, according to the abstract, build a size–latency feasibility envelope for PQ algorithms in a ROS2/DDS-Security handshake for unmanned systems. That setting has a handshake; ours does not. Sikeridis et al. [30] evaluate PQ signatures in the TLS 1.3 handshake (abstract only read). Because two related surveys and the closest competitor were read as abstracts only, we cannot exclude an existing broadcast-focused systematization [TODO: check full text of hyoung2026post and fall2025sok].

III. Method

A. Budget profiles

A profile is a per-message authentication budget B and a frame payload F (Table I). C-V2X uses the 437 B maximum payload of a 10 MHz channel [6]. Remote ID uses the 255 B authentication data field and 23 B of authentication payload per page (25 B frames) [3]. For OSNMA we derive a budget of 16×104 bit $= 208$ B with one 104-bit block per frame; the source reports PQ needs in DSM blocks [4] and our derivation is checked only coarsely (Sec. V). SIB1 uses the 372 B maximum [5]; DNS the 1232 B UDP limit [9]. For C-V2X, SIB1 and DNS a single frame carries the whole budget; the per-fragment header for these is unknown (TODO-RESULT) and set to 0, so fragment counts are lower bounds.

TABLE I
BUDGET PROFILES (BYTES). B: AUTHENTICATION BUDGET PER MESSAGE; F: FRAME PAYLOAD.

Profile	B	F	Source
C-V2X	437	437	[6]
Remote ID	255	23	[3]
OSNMA	208	13	[4] (derived)
5G SIB1	372	372	[5]
DNS (UDP)	1232	1232	[9]

B. Authentication modes

Let s be the maximum signature size of a scheme and p its public key size, both measured. The authentication bytes per message are:

- *full*: $a = s$.
- *sig_cert*: $a = s + (p + s)$, i.e. a certificate holding the subject key and an issuer signature of the same scheme.
- *fragmented*: $a = s + h\,n$ with $n = \lceil s/(F - h) \rceil$ frames and header h.
- *chain*: per message, a MAC tag plus a disclosed key; once per epoch, an anchor of s plus a commitment and timestamp. The mode fits if both the steady-state cost and the anchor fit B. Q1 chain parameters: TODO-RESULT.
- *hybrid*: $a = s_c + s$, with classical partner TODO-RESULT (both verified).

A mode fits a profile if $a \leq B$; we also report a/B and the number of frames $\lceil a/F \rceil$. For *fragmented*, "fits" counts total bytes only; the frame count is the multi-frame cost. The zero-knowledge compression mode of [10] is part of our intended mode set but is not yet evaluated.

C. Measurement

Sizes and sign/verify/keygen times come from liboqs 0.16.0 [TODO: cite: liboqs / Open Quantum Safe] for all signature families it builds (ML-DSA, Falcon, SLH-DSA, MAYO, UOV, SNOVA, CROSS, MQOM), and from OpenSSL via python-cryptography for Ed25519 and ECDSA-P256. Classical timings include Python call overhead and are not comparable with the C liboqs timings. Runs used 1000 iterations per scheme and 20 for CROSS, MQOM and SLH-DSA, on machine `cloud-smoke` (CPU: TODO-RESULT). For the slow families the reported p95/p99 are close to the maximum of 20 samples and are not tail estimates.

D. Loss model (Q2)

We compare two ways to authenticate a SIB1-like broadcast with slot length 160 ms [5]. *Fragmented*: a signature split into k fragments, repeated over $r = 2$ carousel cycles, with deadline rk slots. *Chain*: TESLA-style tags with key delay $d = 2$ slots, epoch $E = 20$ slots, $a = 2$ anchors per epoch and 16 B disclosed keys. In the *cited* scenarios we use $k = 13$

(FN-DSA) and $k = 34$ (ML-DSA) and a 324 B anchor from [5], with 666 B and 2420 B signatures from [4]. In the *oqs* scenarios we use measured Falcon-512 and ML-DSA-44 sizes and an anchor of 454 + 16 + 4 B (measured MAYO-1 signature plus commitment and timestamp); k is still taken from [5], not recomputed. The values of r, d, E, a and the key size are our assumptions, not taken from a source. Loss is i.i.d. per slot at 0–30%; each cell runs 20 seeds of 5000 messages. Metrics: share of messages verified, bytes per verified message, and p95 time to authenticate (over verified messages only).

E. Deadline model (Q3)

For each family we compare the fraction of (parameter set, profile) pairs where the full signature does not fit with the fraction of parameter sets whose sign or verify time exceeds a deadline: 3 ms, the protection timing limit cited in [15], and 100 ms, the basic safety message period [1], [2]. We use p99 where $n = 1000$ and the median where $n = 20$.

IV. Results

A. Sizes and timings

Signature sizes span three orders of magnitude. UOV signatures are 96–260 B but public keys are 412 160–2 869 440 B (66 576 B for compressed OV-Is-pkc). SNOVA ranges from 124 B (public key 9842 B) to 576 B; SNOVA_24_5_4 has a 248 B signature and a 1016 B key. MAYO-2 has a 186 B signature and 4912 B key; MAYO-1 454 B. Falcon-512 signatures are at most 663 B (666 B padded), ML-DSA-44 2420 B, and CROSS, MQOM and SLH-DSA at least 8960, 2916 and 7856 B. The classical references are 64 B (Ed25519) and 72 B (ECDSA-P256).

On the x86 cloud VM, p99 sign times are 253.3 μs for ML-DSA-44, 250.3 μs for Falcon-512, 120.9 μs for MAYO-2 and 85.22 μs for OV-Is. The slowest tails among schemes with 1000 iterations are OV-V-pkc-skc (13 300 μs), SNOVA_60_10_4 (7294 μs) and OV-III-pkc-skc (4053 μs). The slowest SLH-DSA set has a median sign time of 1.282 s.

B. Q1: which (scheme, mode) pairs fit

Fig. 1 and Table II summarize the scoring. A full signature fits in 104 of 1105 (parameter set, profile) pairs: 23 for C-V2X, 17 for Remote ID, 13 for OSNMA, 21 for SIB1 and 30 for DNS. Every fitting set is UOV, SNOVA, MAYO, or (for DNS only) Falcon-512 and Falcon-padded-512. No ML-DSA, SLH-DSA, CROSS or MQOM set fits any profile, even DNS. MAYO-2 is the only MAYO set below the four smaller budgets.

Attaching a certificate removes every option: sig_cert fits in 0 of 1105 pairs, because the schemes with small signatures have public keys of hundreds of bytes to megabytes. Broadcast PQ authentication in these profiles therefore needs keys and certificates distributed out of band or amortized, which is the setting of [5], [6].

The PQ-anchored chain fits in 97 pairs, fewer than the full signature. Its per-message cost is small (Fig. 1), but our fit rule also requires the anchor signature to fit the budget, so the chain inherits the size constraint of its anchor in the smallest profiles (Remote ID, OSNMA). The chain's benefit is per-message cost and loss behaviour (Q2), not a wider set of usable schemes. Hybrid mode loses further sets in C-V2X, Remote ID and OSNMA where the classical signature pushes the total over budget. Fragmentation does not change fit counts, since it adds bytes; it only spreads them over frames.

TABLE II
Fitting parameter sets / sets in family. CROSS, ML-DSA, MQOM and SLH-DSA fit no profile in any mode; sig_cert fits nowhere and is omitted.

Family	Mode	C-V2X	RID	OSNMA	SIB1	DNS
Falcon	full	0/4	0/4	0/4	0/4	2/4
	fragmented	0/4	0/4	0/4	0/4	2/4
	chain	0/4	0/4	0/4	0/4	2/4
	hybrid	0/4	0/4	0/4	0/4	2/4
MAYO	full	1/4	1/4	1/4	1/4	4/4
	fragmented	1/4	1/4	1/4	1/4	4/4
	chain	1/4	1/4	1/4	1/4	4/4
	hybrid	1/4	1/4	0/4	1/4	4/4
SNOVA	full	10/12	7/12	3/12	8/12	12/12
	fragmented	10/12	7/12	3/12	8/12	12/12
	chain	10/12	3/12	3/12	8/12	12/12
	hybrid	8/12	3/12	1/12	8/12	12/12
UOV	full	12/12	9/12	9/12	12/12	12/12
	fragmented	12/12	9/12	9/12	12/12	12/12
	chain	12/12	9/12	6/12	12/12	12/12
	hybrid	12/12	6/12	6/12	12/12	12/12

C. Q2: chain vs. fragmented under loss

Without loss, the chain costs 64.4 B per verified message in the cited scenarios (79.4 B with the measured MAYO-1 anchor), against 1332 B and 4840 B for fragmented FN-DSA and ML-DSA; p95 time is 480 ms against 2080 ms and 5440 ms (Fig. 2).

At 10% loss the chain verifies 0.8935 of messages at 72.08 B each, versus 0.878 at 1517 B for 13-fragment FN-DSA, and 0.8907 at 72.31 B versus 0.7117 at 6801 B for 34-fragment ML-DSA. At 30% loss the gap widens: 0.637 versus 0.2932 (FN-DSA) and 0.6344 versus 0.04019 (ML-DSA), at 101.5 B versus 120 900 B per verified ML-DSA message.

The result is not uniform. Counting cells with loss > 0 where the chain is better by more than two standard deviations, it wins on bytes per verified message in 24/24 cells and on p95 time in 24/24, but on verified share in only 22/24. At 5% loss with 13 fragments, fragmented signatures verify a larger share than the chain (0.9685 versus 0.9485). We have not isolated the cause; a plausible one is that a chain message also fails when its disclosed key or anchor is lost. In a one-at-a-time sweep of r, d, E and a at 5, 10 and 30% loss, the chain is better on share in 110/144 cells and on bytes and p95 in 142/144 each. The p95 comparison also flatters fragmentation: p95 is taken over verified messages only and is capped by the deadline (4160 ms and 10 880 ms), so a low verified share hides long waits.

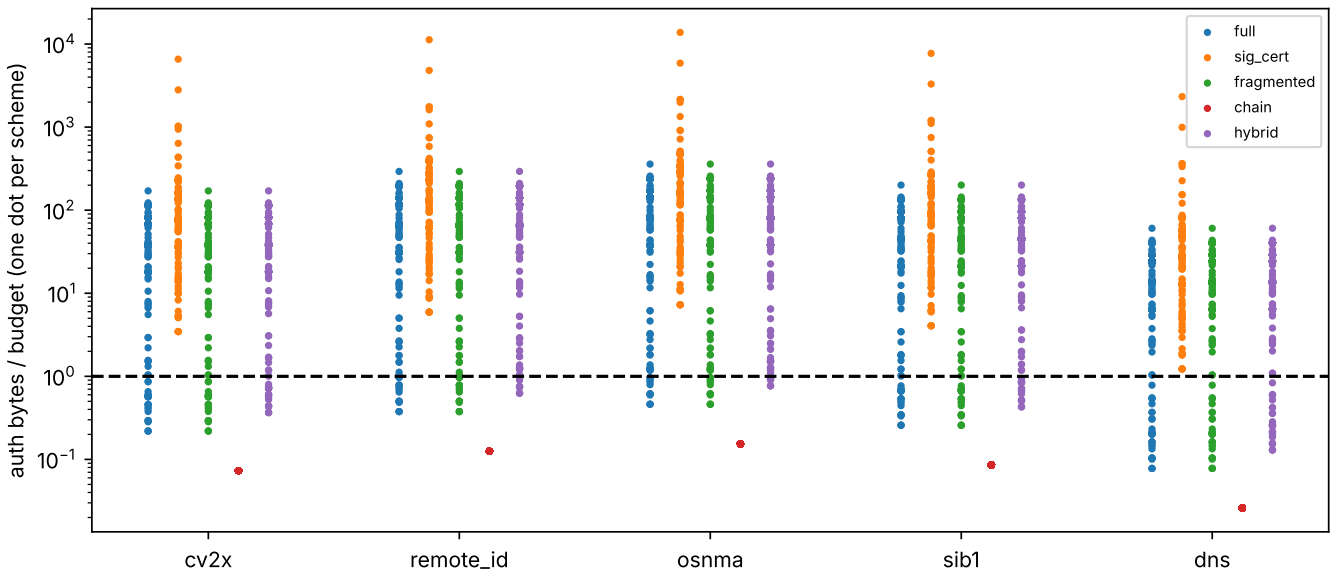


Fig. 1. Authentication bytes divided by budget, one dot per PQ parameter set, per profile and mode (log scale). Points below the dashed line fit. Provisional smoke-test data.

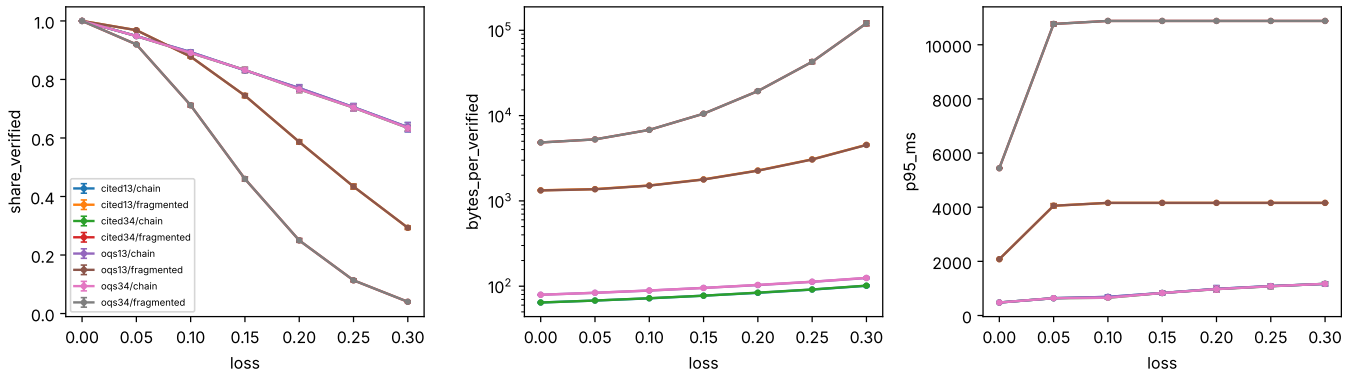


Fig. 2. Loss model (Q2): share of messages verified, bytes per verified message (log), and p95 time to authenticate (ms) versus i.i.d. loss. Mean ± std over 20 seeds. Chain parameters are assumed; see Sec. III. Provisional.

We therefore do not claim that the chain dominates. Under these smoke assumptions it is clearly cheaper in bytes and faster, and it verifies more messages except at low loss with few fragments. We could not reproduce the 10%-loss point of [5] (about 41 B versus 1971 B retransmission overhead), because that work uses a retransmission accounting our carousel model does not implement (TODO-RESULT).

D. Q3: is size or the timing tail binding?

Size misfit exceeds the deadline-miss fraction at both deadlines in 6 of 8 families (Table III, Fig. 3); no family is timing-bound at both deadlines. Two families qualify this. SLH-DSA is a tie at 3 ms: every set misses both the budgets and the 3 ms deadline. For UOV the 3 ms miss fraction (0.167) is larger than the size misfit (0.10); the misses come from the pkc-skc variants with sign tails of several milliseconds. For a 3 ms application such as sampled-value protection, UOV parameter set choice is thus timing-constrained, not size-constrained.

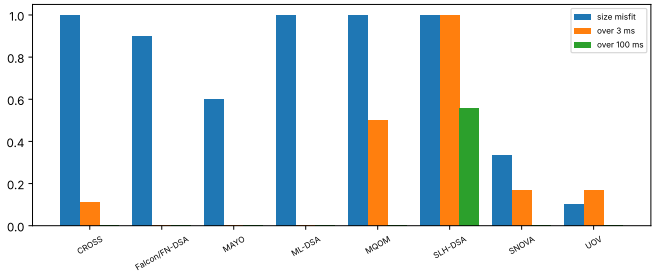


Fig. 3. Per family: fraction of (set, profile) pairs where the full signature does not fit, and fraction of sets whose sign or verify tail exceeds 3 ms or 100 ms. x86 cloud VM only. Provisional.

TABLE III
SIZE MISFIT VERSUS DEADLINE MISS PER FAMILY (X86). †MEDIAN OF 20 SAMPLES; TAIL NOT ESTIMABLE.

Family	Sets	Basis	Misfit	>3 ms	>100 ms
CROSS	18	median†	1.00	0.111	0
Falcon/FN-DSA	4	p99	0.90	0	0
MAYO	4	p99	0.60	0	0
ML-DSA	3	p99	1.00	0	0
MQOM	12	median†	1.00	0.500	0
SLH-DSA	156	median†	1.00	1.000	0.558
SNOVA	12	p99	0.333	0.167	0
UOV	12	p99	0.10	0.167	0

This answer holds on x86 only. On microcontrollers, timings may be much larger [TODO: cite or measure: Cortex-M sign/verify for UOV, SNOVA, MAYO; chhetri2026benchmarking covers ML-DSA only], which could make timing binding; we did not measure this.

V. LIMITATIONS

- **Provisional data.** All numbers are smoke-test runs on a shared cloud VM (CPU: TODO-RESULT) with 1000 iterations (20 for CROSS, MQOM, SLH-DSA). Publication runs with 10 000 and 200 iterations on target machines are pending.
- **Slow-family tails.** For CROSS, MQOM and SLH-DSA, p95/p99 are near-maxima of 20 samples and Q3 uses medians for them.
- **Q2 is inconclusive.** Results depend on assumed r, d, E, a and key size, on i.i.d. loss (no burst loss), and on fragment counts k taken from [5] rather than recomputed from each profile's payload.
- **Q3 is x86 only.** MCU timing is not measured; the conclusion does not cover constrained devices.
- **Incomplete profiles.** Fragment headers for C-V2X, SIB1 and DNS are unknown (TODO-RESULT, TODO-RESULT, TODO-RESULT) and set to 0. The OSNMA budget is derived; the 67/71 DSM-block figures of [4] are not reproduced (TODO-RESULT), and our block arithmetic does not match them without the full text. The hybrid partner (TODO-RESULT) and the Q1 chain parameters (TODO-RESULT) are not yet recorded in the result summary.
- **Baselines not reproduced.** The 34/13 SIB1 fragments and 324 B overhead of [5] (TODO-RESULT) and its 10%-loss point (TODO-RESULT); the ZK compression of [10] (TODO-RESULT); the DNS transfer sizes of [9] (TODO-RESULT); and the size tables of [8].
- **Scope.** Stateful hash-based signatures (LMS/XMSS) are not benchmarked. ETSI ITS, ADS-B, AIS and LoRaWAN profiles are not included. Security of the modes (e.g. hybrid downgrade, time synchronization attacks on the chain) is not analysed.
- **Related work.** Several related papers, including the closest competitor [29], were read as abstracts only.
- **Bookkeeping.** The result summary reports 168 SLH-DSA parameter sets in one place and 156 in the family table [TODO: check SLH-DSA set count].

VI. CONCLUSION

Scoring one set of authentication modes against five handshake-free broadcast budgets shows, in provisional x86 data, that only UOV, SNOVA, MAYO and two Falcon parameter sets fit any profile with a full signature, that no set fits with a same-scheme certificate, and that size rather than x86 timing is usually the binding constraint. A PQ-anchored key chain is far cheaper per verified message than fragmented signatures under loss, but it does not verify more messages at low loss and the comparison rests on assumed parameters. Next steps are publication runs on target hardware including microcontrollers, recomputed fragment counts, burst loss, and reproduction of the [5] and [4] baselines.

ARTIFACT

Scripts that regenerate every number in this paper from raw CSVs: [TODO: artifact URL].

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